

Homework 4: FFT, Sampling, and Coupled Oscillators

Part 1: Before running anything

1. An FFT, or Fast Fourier Transform, tells us how strongly different frequency components are present in a sampled signal. In other words, it converts information from the time domain into the frequency domain.
2. The sampling rate f_s is the number of samples taken per second. The Nyquist frequency is half of the sampling rate,

$$f_N = \frac{f_s}{2},$$

and it is the highest frequency that can be represented without aliasing. The frequency resolution is the spacing between neighboring FFT frequency bins,

$$\Delta f = \frac{1}{T} = \frac{f_s}{N},$$

where T is the total acquisition time and N is the number of samples.

3. A uniformly spaced time grid is important because the standard FFT assumes that samples are separated by a constant time interval. If the samples are not equally spaced, the FFT bin frequencies no longer correspond correctly to the data.

Part 2: Build and run

1. After running

`make run`

the project creates the following CSV files in the `output/` directory:

- `output/good_sampling_signal.csv`
- `output/good_sampling_spectrum.csv`
- `output/undersampled_signal.csv`
- `output/undersampled_spectrum.csv`
- `output/short_record_signal.csv`
- `output/short_record_spectrum.csv`
- `output/coupled_oscillators_time.csv`
- `output/coupled_oscillators_spectrum.csv`

2. The program that studies pure sampling issues is `src/sampling_demo.c`.
3. The program that studies a mechanics problem is `src/coupled_oscillators_fft.c`.
4. In the well-sampled case, the two reported frequencies are approximately 50 Hz and 120 Hz.

Part 3: Sampling and aliasing

1. In the undersampled case, the sampling frequency is

$$f_s = 128 \text{ Hz}.$$

2. Therefore the Nyquist frequency is

$$f_N = \frac{f_s}{2} = \frac{128 \text{ Hz}}{2} = 64 \text{ Hz}.$$

3. A 120 Hz component cannot be reconstructed faithfully because 120 Hz is larger than the Nyquist frequency 64 Hz. Frequencies above Nyquist fold back into lower apparent frequencies after sampling.

4. The program reports an aliased peak near 8 Hz. Mathematically, sampled sinusoids can become indistinguishable when their frequencies differ by integer multiples of the sampling frequency. For the 120 Hz component sampled at 128 Hz, the alias is

$$f_{\text{alias}} = |120 - 128| \text{ Hz} = 8 \text{ Hz}.$$

5. Physically, the sampling is too slow to follow the rapid oscillation. The discrete samples then look like they came from a much slower oscillation, producing a false low-frequency peak in the FFT spectrum.

Part 4: Frequency resolution

1. In the short-record case, the signal contains 50 Hz and 55 Hz.
2. From the given setup,

$$T = \frac{N}{f_s} = \frac{64}{512} \text{ s} = 0.125 \text{ s}.$$

3. Thus the FFT frequency spacing is

$$\Delta f = \frac{1}{T} = \frac{1}{0.125 \text{ s}} = 8 \text{ Hz}.$$

4. The two frequencies differ by only 5 Hz, which is smaller than the FFT bin spacing of 8 Hz. Therefore the FFT cannot cleanly separate the two components.

5. To improve the separation, the first quantity to change should be the total acquisition time. Increasing T decreases $\Delta f = 1/T$, producing finer frequency bins. Changing the plotting tool or output file format does not improve the underlying frequency resolution.

Part 5: Coupled oscillators

1. The equations of motion used in the code are

$$m\ddot{x}_1 = -(k + k_c)x_1 + k_c x_2,$$

$$m\ddot{x}_2 = k_c x_1 - (k + k_c)x_2,$$

where m is the mass, k is the wall-spring constant, k_c is the coupling spring constant, and x_1, x_2 are the displacements of the two masses.

2. These equations represent two identical masses attached to fixed walls by springs and coupled to each other by another spring.

3. The two normal modes are:

- an in-phase mode, where both masses move in the same direction;
- an out-of-phase mode, where the masses move in opposite directions.

4. The FFT of $x_1(t)$ shows more than one important frequency because the chosen initial condition excites a combination of normal modes rather than a single pure mode.

The theoretical normal-mode frequencies are

$$f_{\text{in}} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}, \quad f_{\text{out}} = \frac{1}{2\pi} \sqrt{\frac{k + 2k_c}{m}}.$$

5. Comparing the numerical FFT peaks with these theoretical values is useful because it checks whether the numerical simulation is consistent with the analytical physics. It also helps detect coding or modeling mistakes.

Part 6: Plot inspection

1. The plot that makes aliasing easiest to see is the sampling spectrum plot, especially the undersampled spectrum. In that plot, the high-frequency 120 Hz component appears as a false low-frequency peak near 8 Hz.

2. The plot that makes limited frequency resolution easiest to see is the short-record spectrum. The peaks corresponding to 50 Hz and 55 Hz are difficult to distinguish because the FFT bin spacing is too coarse.

3. In the coupled-oscillator FFT plot, the numerical peaks line up closely with the theoretical reference frequencies.

4. If they are not exactly identical, the small difference can be explained by finite observation time, discrete FFT bin spacing, numerical solver tolerance, and the fact that the exact physical frequency may not fall exactly on a discrete FFT bin.

Part 7: Small code modification

1. I choose Option B. In the short-record case, I increase the number of samples while keeping the time step, or equivalently the sampling interpretation, clear.

2. Since

$$T = N\Delta t,$$

increasing N increases the total observation time T .

Because the frequency spacing is

$$\Delta f = \frac{1}{T},$$

a larger T gives a smaller Δf . Therefore, before rerunning the code, I predict that the 50 Hz and 55 Hz peaks should become more distinguishable.

3. After rerunning, the FFT output should show better separation between the two nearby peaks. The improvement happens because the longer record gives the FFT a finer frequency grid.

Part 8: Reflection

From this project, I learned that the FFT is not just a button that finds frequencies automatically. Its result depends strongly on the sampling rate, the Nyquist limit, and the total observation time. In particular, undersampling can make a high-frequency signal appear as a false low-frequency signal, while a short observation time can make nearby frequencies hard to distinguish.